**Software Requirements Specification (SRS) – ScrollControl**

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## 1. Introduction

### 1.1 Purpose

This document provides a detailed description of the functional and non-functional requirements for the ScrollControl application. It serves as a guide for developers, designers, and stakeholders throughout the development lifecycle.

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### 1.2 Scope

ScrollControl is a mobile application designed to help users reduce excessive scrolling by:

- Tracking app usage
- Providing real-time reminders
- Enabling focus sessions

The system will primarily target Android devices, with limited functionality on iOS and optional support via a web dashboard.

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### 1.3 Definitions

- **Focus Session**: A timed period during which the user avoids distractions
- **App Usage Tracking**: Monitoring time spent on selected applications
- **Smart Interrupt**: A notification triggered after prolonged app usage

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## 2. Overall Description

### 2.1 Product Perspective

ScrollControl is a standalone application that:

- Runs on mobile devices
- Uses system-level permissions for tracking usage
- Optionally connects to a backend for data sync

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### 2.2 Product Functions

- Start and manage focus sessions
- Track application usage
- Notify users after excessive scrolling
- Display usage statistics

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### 2.3 User Characteristics

- Basic smartphone users
- Familiar with mobile apps
- No technical expertise required

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### 2.4 Constraints

- Limited access to app usage data on iOS
- Strict permission policies on Android
- Must comply with privacy regulations

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### 2.5 Assumptions

- Users will grant required permissions
- Users want to improve productivity
- Device supports required APIs

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## 3. System Features & Functional Requirements

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### 3.1 Focus Timer

**Description**

Allows users to initiate and complete focus sessions.

**Functional Requirements**

- FR1.1: System shall allow user to start a focus session

- FR1.2: System shall allow user to stop a focus session
- FR1.3: System shall display countdown timer
- FR1.4: System shall run timer in background
- FR1.5: System shall store completed sessions

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**3.2 App Usage Tracking (Android Only)**

**Description**

Tracks time spent on selected apps.

**Functional Requirements**

- FR2.1: System shall request usage access permission
- FR2.2: System shall track app usage duration
- FR2.3: System shall store usage data locally
- FR2.4: System shall display usage per app

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**3.3 Smart Interrupt**

**Description**

Notifies users when prolonged scrolling is detected.

**Functional Requirements**

- FR3.1: System shall detect continuous usage beyond threshold
- FR3.2: System shall trigger notification
- FR3.3: Notification shall display usage duration
- FR3.4: Notification shall provide actions:
    - Continue
    - Start focus session

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**3.4 Daily Statistics**

**Description**

Displays insights into user behavior.

**Functional Requirements**

- FR4.1: System shall calculate total daily usage
- FR4.2: System shall identify most used apps
- FR4.3: System shall display focus time
- FR4.4: System shall present data in simple visual format

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**3.5 Permissions Management**

**Description**

Handles user permissions transparently.

**Functional Requirements**

- FR5.1: System shall explain permission usage before request
- FR5.2: System shall request necessary permissions only
- FR5.3: System shall allow users to revoke permissions
- FR5.4: System shall function partially without permissions

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**4. External Interface Requirements**

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**4.1 User Interface**

- Minimal, dark-themed design
- Large timer display
- Simple navigation (max 3 tabs)

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**4.2 Hardware Interface**

- Android smartphone
- Optional: iOS device (limited features)

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**4.3 Software Interface**

- Android Usage Stats API

- Notification system
- Optional backend API (Laravel)

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**5. Non-Functional Requirements**

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**5.1 Performance**

- App response time ≤ 2 seconds
- Efficient background processing

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**5.2 Usability**

- Intuitive navigation
- Maximum 2–3 steps per main action
- Clear and readable UI

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**5.3 Reliability**

- App shall not crash during tracking
- Data shall persist across sessions

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**5.4 Security & Privacy**

- Data stored locally by default
- No data shared without user consent
- Transparent data usage

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**5.5 Maintainability**

- Modular code structure
- Easy to update features

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**5.6 Scalability**

- Backend support for future expansion
- Ability to add new features without redesign

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**6. System Architecture**

**6.1 Overview**

The system consists of:

- Frontend (Flutter app)
- Local storage (SQLite)
- Optional backend (Laravel API)

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**6.2 Data Flow**

1. User interacts with app
2. App collects usage data
3. Data stored locally
4. Optional sync with backend

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**7. Data Requirements**

**Entities**

**User**

- id
- email (optional)
- created_at

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**Focus Session**

- id
- user_id
- duration
- date

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**App Usage**

- id
- app_name
- usage_time
- date

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**8. Constraints**

- Must comply with Google Play policies
- Must minimize battery usage
- Must avoid misuse of restricted APIs

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**9. Assumptions & Dependencies**

- Android OS provides accurate usage data
- Notification system functions correctly
- User grants required permissions

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**10. Acceptance Criteria**

The system is considered complete when:

- Users can track app usage
- Focus sessions function correctly
- Notifications trigger as expected
- Data is displayed accurately
- App complies with store policies

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